Regression Techniques

Wednesday, September 30, 2026

1 Leave-One-Out Diagnostics

1.1 Key Result: Change in the OLS Estimator

Result:

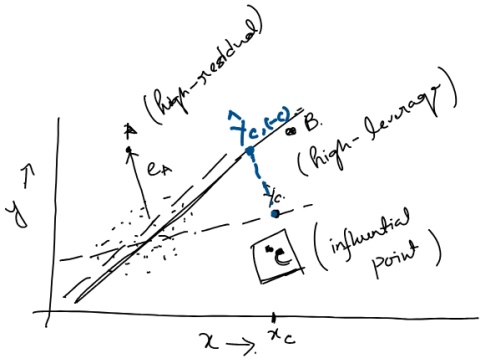
$$\underbrace{\hat{\boldsymbol{\beta}}_{OLS} - \hat{\boldsymbol{\beta}}_{OLS,(-i)}}_{DF\beta} = (X^T X)^{-1}\mathbf{x}_{(i)}\left(\frac{e_i}{1-p_{ii}}\right)$$

Equivalently,

$$\hat{\boldsymbol{\beta}}_{OLS,(-i)} = \hat{\boldsymbol{\beta}}_{OLS} - (X^T X)^{-1}\mathbf{x}_{(i)}\left(\frac{e_i}{1-p_{ii}}\right)$$

where the leverage term satisfies

$$p_{ii} = \frac{1}{n} + \frac{1}{n}MD_i^2.$$



1.2 DFBETA for the j-th Predictor

Let $DF\beta_j$ denote the change restricted to the j-th predictor. Then

$$\underbrace{\hat{\beta}_{OLS,j} - \hat{\beta}_{OLS,(-i),j}}_{DF\beta_j} = \underbrace{(0,\ldots,0,1_j,0,\ldots,0)}_{e_j}(X^T X)^{-1}\mathbf{x}_{(i)}\left(\frac{e_i}{1-p_{ii}}\right) \tag{1}$$

1.3 DFBETAS ($DF\beta S$)

Equation (1) is standardized by

$$\text{var}\left(\hat{\beta}_{OLS,j}\right) = \left[(j,j)\text{-th component of var}(\hat{\boldsymbol{\beta}}_{OLS})\right] = \hat{\sigma}^2\left[(X^T X)^{-1}\right]_{j,j}$$

where

$$\hat{\sigma}^2 = \hat{\sigma}^2_{UE,(-i)}.$$

2 Derivation of $RSS_{(-i)}$

$$RSS_{(-i)} = \sum_{\substack{k=1\\k\neq i}}^{n} \left(Y_k - \hat{Y}_{k,(-i)}\right)^2 = \sum_{\substack{k=1\\k\neq i}}^{n} \left(Y_k - \mathbf{x}^T_{(k)}\hat{\boldsymbol{\beta}}_{OLS,(-i)}\right)^2$$

$$= \sum_{\substack{k=1\\k\neq i}}^{n} \left(Y_k - \mathbf{x}^T_{(k)}\left[\hat{\boldsymbol{\beta}}_{OLS} - (X^TX)^{-1}\mathbf{x}_{(i)}\frac{e_i}{1-p_{ii}}\right]\right)^2$$

$$= \sum_{\substack{k=1\\k\neq i}}^{n} \left(\underbrace{Y_k - \hat{Y}_k}_{e_k} + p_{ik}\frac{e_i}{1-p_{ii}}\right)^2$$

$$= \sum_{k=1}^{n} \left(e_k + \frac{p_{ik}}{1-p_{ii}}e_i\right)^2 - \left(e_i + p_{ii}\frac{e_i}{1-p_{ii}}\right)^2$$

$$= \underbrace{\sum_{k=1}^{n} e_k^2}_{RSS} + 2\frac{e_i}{1-p_{ii}}\underbrace{\sum_{k=1}^{n} p_{ik}e_k}_{0} + \frac{e_i^2}{(1-p_{ii})^2}\underbrace{\sum_{k=1}^{n} p_{ik}^2}_{p_{ii}} - \frac{1}{(1-p_{ii})^2}e_i^2$$

Notation.

$$X = [\mathbf{1}\ \ \mathbf{x}_1\ \ \cdots\ \ \mathbf{x}_p] = \begin{bmatrix}\mathbf{x}^T_{(1)}\\ \vdots \\ \mathbf{x}^T_{(n)}\end{bmatrix}, \qquad \mathbf{x}^T_{(k)}(X^TX)^{-1}\mathbf{x}_{(i)} = p_{ki} = p_{ik}.$$

2.1 Claim and Proof

Claim: $\sum_{k=1}^{n} p_{ik}e_k = 0.$

Proof: $P(X)e = P(X)\big[I - P(X)\big]Y = 0$, and $\sum_{k=1}^{n} p_{ik}e_k$ is the i-th row of $P(X)e$.

2.2 Variance Estimate

$$RSS_{(i)} = RSS - \frac{e_i^2}{(1-p_{ii})^2}(1-p_{ii}) = RSS - \frac{e_i^2}{1-p_{ii}}$$

$$\hat{\sigma}^2_{UE,(-i)} = \frac{RSS_{(-i)}}{n-p-2} = \hat{\sigma}^2_{UE}\left(\frac{n-p-1}{n-p-2}\right) - \frac{e_i^2}{(n-p-2)(1-p_{ii})} \qquad (*)$$

$$DF\beta S_j = \frac{\hat{\beta}_{OLS,j} - \hat{\beta}_{OLS,(-i),j}}{\hat{\sigma}_{UE,(-i)}\left[(X^TX)^{-1}\right]^{1/2}_{j,j}}$$

Compare:

$$|DF\beta S_j| \leq \frac{2}{\sqrt{n}}$$

3 DF FIT

$$DFFIT = \hat{Y}_i - \hat{Y}_{i(-i)}$$

$$DFFit = \mathbf{x}_{(i)}^T \hat{\boldsymbol{\beta}}_{OLS} - \mathbf{x}_{(i)}^T \hat{\boldsymbol{\beta}}_{OLS,(-i)} = \mathbf{x}_{(i)}^T (X^T X)^{-1} \mathbf{x}_{(i)} \frac{e_i}{1-p_{ii}} = \frac{p_{ii}}{1-p_{ii}} e_i$$

3.1 DFFITS (Standardized Version)

DFFITS standardizes by $\sqrt{\text{var}(\hat{Y}_i)} = \hat{\sigma}\, p_{ii}^{1/2}$, with $\hat{\sigma}_{UE,(-i)}$ used as an estimate.

$$DFFITS = \frac{p_{ii}}{1-p_{ii}} \cdot \frac{e_i}{\hat{\sigma}_{UE,(-i)}\, p_{ii}^{1/2}} = \left(\frac{p_{ii}^{1/2}}{\sqrt{1-p_{ii}}}\right) \cdot \underbrace{\frac{e_i}{\hat{\sigma}_{UE,(-i)}\sqrt{1-p_{ii}}}}_{\substack{t_i \\ \text{(standardized residual)} \\ \sim\ t\text{-distribution}}}$$

with $|t_i| \leq 2$.

Approximation for $p \ll n$.

$$p_{ii} \sim \frac{p}{n}, \qquad (1-p_{ii}) \approx 1, \qquad \sqrt{\frac{p_{ii}}{1-p_{ii}}} \sim \sqrt{\frac{p}{n}}.$$

Thumb Rule:

$$|DFFITS| \leq 2\sqrt{\frac{p}{n}}$$

4 Covariance Ratio

$$\frac{\det\left(\text{var}\left(\hat{\boldsymbol{\beta}}_{OLS,(-i)}\right)\right)}{\det\left(\text{var}\left(\hat{\boldsymbol{\beta}}_{OLS}\right)\right)} = \text{Cov-Ratio}$$

$$\text{Cov-Ratio} = \frac{\det\left(\hat{\sigma}^2_{UE,(-i)}\left(X^T_{(-i)}X_{(-i)}\right)^{-1}\right)}{\det\left(\hat{\sigma}^2_{UE}\left(X^T X\right)^{-1}\right)} = \frac{\hat{\sigma}^2_{UE,(-i)}}{\hat{\sigma}^2_{UE}} \cdot \frac{\det(X^T X)}{\det\left(X^T_{(-i)}X_{(-i)}\right)}$$

4.1 Determinant Identity

$$X^T X = X^T_{(-i)}X_{(-i)} + \mathbf{x}_{(i)}\mathbf{x}^T_{(i)}$$

$$\left|X^T_{(-i)}X_{(-i)}\right| = \left|X^T X - \mathbf{x}_{(i)}\mathbf{x}^T_{(i)}\right| = |X^T X| \left|I - \underbrace{(X^T X)^{-1}\mathbf{x}_{(i)}\mathbf{x}^T_{(i)}}_{A}\right|$$

where $I - A$ is a **rank-1 matrix**.

$$\lambda_j(I+A) = 1 + \lambda_j(A), \qquad \big[(I+A)\lambda = \lambda + A\lambda\big]$$

Here $(X^TX)^{-1}\mathbf{x}_{(i)}\mathbf{x}_{(i)}^T$ has only one non-zero eigenvalue, using

$$ab^T = \begin{bmatrix} a_1 b^T \\ \vdots \\ a_{p+1} b^T \end{bmatrix}.$$

Non-zero eigenvalue.

$$\mathbf{x}_{(i)}^T (X^TX)^{-1} \mathbf{x}_{(i)} = p_{ii}$$

$$|I + A| = \big(1 + \lambda_{\max}(A)\big) \cdot 1 \cdots 1$$

Recall: The non-zero eigenvalues of AB and BA are the same. Hence the non-zero eigenvalue of ab^T is the same as that of b^Ta.

Thus,

$$\left| X_{(-i)}^T X_{(-i)} \right| = |X^TX|\,(1 - p_{ii})$$

$$\text{Cov-Ratio} = \frac{\hat\sigma^2_{UE,(-i)}}{\hat\sigma^2_{UE}} \cdot \frac{|X^TX|}{|X^TX|(1-p_{ii})} = \frac{\hat\sigma^2_{UE}\left(\dfrac{n-p-1}{n-p-2}\right) - \dfrac{e_i^2}{(n-p-2)(1-p_{ii})}}{\hat\sigma^2_{UE}} \cdot \frac{1}{1-p_{ii}}$$

$$\approx \frac{1}{1-p_{ii}}\left(1 - \frac{e_i^2}{(n-p-2)\hat\sigma^2_{UE}(1-p_{ii})}\right) = \frac{1}{1-p_{ii}}\left(1 - \frac{r_i^2}{n-p-2}\right)$$

where r_i^2 is the **Studentized residual**.

$$|\text{Cov-Ratio} - 1| \le \frac{3p}{n}$$

Interpretation.

- **High** Cov-Ratio: $p_{ii} \approx 1$ (high-leverage point), e_i^2 small.
- **Low** Cov-Ratio: e_i^2 or r_i^2 are large.

5 Cook's Distance

$$\frac{1}{\sigma^2}\left(\boldsymbol\beta - \hat{\boldsymbol\beta}_{OLS}\right)^T (X^TX)\left(\boldsymbol\beta - \hat{\boldsymbol\beta}_{OLS}\right) \sim \chi^2_{p+1}$$

$$\frac{RSS}{\sigma^2} \sim \chi^2_{(n-p-1)} \quad \text{independently.}$$

$$\frac{1}{p\,\hat\sigma^2_{UE}}\left(\boldsymbol\beta - \hat{\boldsymbol\beta}_{OLS}\right)^T (X^TX)\left(\boldsymbol\beta - \hat{\boldsymbol\beta}_{OLS}\right) \sim F_{p+1,\,n-p-1}$$

$$P_\beta\left(\left\{\boldsymbol\beta : \frac{1}{(1+p)\hat\sigma^2_{UE}}\left(\boldsymbol\beta - \hat{\boldsymbol\beta}_{OLS}\right)^T (X^TX)\left(\boldsymbol\beta - \hat{\boldsymbol\beta}_{OLS}\right) \le F^{\alpha}_{p+1,\,n-p-1}\right\}\right) = (1-\alpha)$$

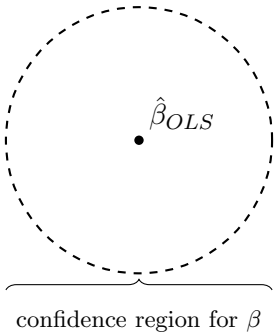


Figure 1: The $(1-\alpha)$ confidence region for β centered at $\hat{\beta}_{OLS}$.

5.1 Cook's D Formula

$$\text{Cook's D} = \frac{\left(\hat{\boldsymbol{\beta}}_{OLS,(-i)} - \hat{\boldsymbol{\beta}}_{OLS}\right)^T X^T X \left(\hat{\boldsymbol{\beta}}_{OLS,(-i)} - \hat{\boldsymbol{\beta}}_{OLS}\right)}{(1+p)\,\hat{\sigma}^2_{UE}}$$

If Cook's D $\leq F^{0.1}_{p+1,\,n-p-1}$, **then** $\mathbf{x}_{(i)}$ is an usual observation.

$$= \frac{\mathbf{x}^T_{(i)}(X^TX)^{-1}\mathbf{x}_{(i)}}{(p+1)\,\hat{\sigma}^2_{UE}} \cdot \frac{e_i^2}{(1-p_{ii})^2}$$

$$= \frac{p_{ii}}{(p+1)(1-p_{ii})^2} \cdot \frac{e_i^2}{\hat{\sigma}^2_{UE}}$$

$$= \frac{p_{ii}}{1-p_{ii}} \cdot \frac{1}{1+p} \cdot r_i^2$$

$$r_i = \frac{e_i}{\sqrt{1-p_{ii}}\,\hat{\sigma}_{UE}} \qquad t_i = \frac{e_i}{\hat{\sigma}_{UE,(-i)}\sqrt{1-p_{ii}}}$$